

Discontinuity-Free Decision Support with Quantitative Argumentation Debates

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Abstract

IBIS (Issue Based Information System) provides a widely adopted approach for knowledge representation especially suitable for the challenging task of representing *wicked* decision problems. While many tools for visualisation and collaborative development of IBIS graphs are available, automated decision support in this context is still underdeveloped, even though it would benefit several applications. *QuAD* (Quantitative Argumentation Debate) frameworks are a recently proposed IBIS-based formalism encompassing automated decision support by means of an algorithm for quantifying the strength of alternative decision options, based on aggregation of the strength of their attacking and supporting arguments. The initially proposed aggregation method, however, may give rise to discontinuities. In this paper we propose a novel, discontinuity-free algorithm for computing the strength of decision options in *QuAD* frameworks. We prove that this algorithm features several desirable properties and we compare the two aggregation methods, showing that both may be appropriate in the context of different application scenarios.

1 Introduction

The representation of decision problems is an extremely challenging task, especially when these problems are *wicked*, namely they rely on incomplete or conflicting knowledge, they involve several stakeholders with conflicting views and they are closely linked with other decision problems (Churchman 1967). Instances of wicked decision problems occur naturally, for example, in engineering design (Marashi and Davis 2006) and e-democracy (Loukis, Xenakis, and Tseperli 2009; Gordon and Richter 2002). *IBIS* (Issue Based Information System) (Kunz and Rittel 1970) provides a powerful and widely adopted approach for knowledge representation especially suitable for wicked decision problems. While many tools for visualisation and collaborative development of IBIS graphs are available (Conklin et al. 2001; Bracewell et al. 2009; Aurisicchio and Bracewell 2013), automated decision support in this context is still underdeveloped, even though it would benefit several applications. *QuAD* (Quantitative Argumentation Debate) frameworks (Baroni et al. 2015) are a recently proposed IBIS-based formalism encompassing automated decision support by means of an algorithm for computing the

strength of answers to decision questions, given an IBIS graph of a restricted kind (see (Baroni et al. 2015) for details). The calculation aggregates the strength of attacking and supporting arguments for the answers and for other opinions, recursively. The *QuAD* algorithm has been integrated within the *designVUE* tool¹ (Baroni et al. 2015) and the *Arg&Dec* system² (Aurisicchio et al. 2015) and has proven useful in several applications in engineering design.

Figure 1 shows an IBIS graph/*QuAD* framework as visualized in *Arg&Dec*. The graph is an abstraction of a real

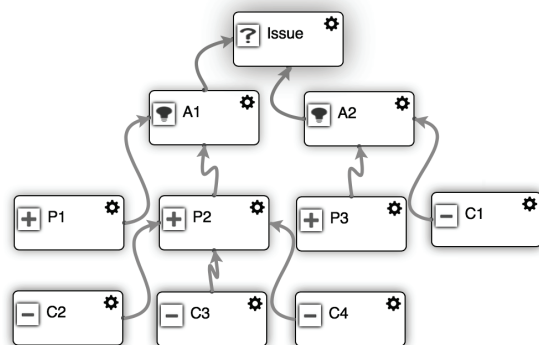


Figure 1: Example IBIS graph

decision-making problem in design engineering, where the *Issue* node represents the question of how to control the ventilation in a room; the *Answer* nodes A1 and A2 are the available options, with A1=“building management system control” and A2=“user control”; the two *Pro Argument* nodes P1 and P2 support A1 and P3 supports A2, with P1=“energy is saved”, P2=“elderly occupants require more simple settings” and P3=“user satisfaction is increased”; the *Con Argument* nodes C2, C3 and C4 attack P2 and C1 attacks A2, with C1=“user negligence can lead to losses”, C2=“user control systems are relatively simple”, C3=“elderly occupants who have difficulty will probably have carers” and C4=“the building’s target market is young professionals”.

The same IBIS graph may be an abstraction of an e-democracy decision problem, as discussed in Section 6.

¹ www3.imperial.ac.uk/designengineering/tools/designvue

² www.arganddec.com

QuAD frameworks admit Abstract Argumentation frameworks (Dung 1995) as an instance (Baroni et al. 2015), and the QuAD algorithm replaces standard semantic notions of acceptability, as in (Dung 1995), with a quantitative notion of strength, starting from a *base score* (or intrinsic strength) of answers and arguments. For example, assuming that each answer and argument in Figure 1 has a medium base score of 0.5, the QuAD algorithm determines, in particular, that A1 has strength 0.765625 and is thus stronger than A2, with strength 0.5. The higher strength of A1 reflects a higher level of support (albeit mitigated by the bottom layer of attacks).

As noted in (Baroni et al. 2015), the aggregation of opinions used in the QuAD algorithm causes a form of *discontinuity* of the notion of strength: for example, an opinion with very strong attackers and no supporter may have a strength of 0 (the lowest possible), but adding even a single very weak supporter for the opinion may increase its strength abruptly to a (comparatively) high value; dually, the strength of an opinion with very strong supporters and no attackers may jump from 1 (the highest possible value) to a (comparatively) low value. As discussed in (Baroni et al. 2015), this behaviour may be suitable in some contexts but not in others.

In this paper we propose a novel, discontinuity-free algorithm, called *DF-QuAD*, for computing the strength of opinions in QuAD frameworks. We also prove several desirable properties of this algorithm, also exhibited by the original QuAD algorithm (Baroni et al. 2015), and, in addition, identify and prove a novel “discontinuity-freeness” property, not exhibited by the original QuAD algorithm. Furthermore, we identify special classes of QuAD frameworks for which the QuAD algorithm and the DF-QuAD algorithm coincide. We then compare the two algorithms in the context of two different application scenarios, engineering design and e-democracy, showing that both may be beneficial depending on the characteristics of the scenario. Finally, we show how the DF-QuAD algorithm may be used in a *reverse engineering* context, where recommendations are interactively given to users as to how to modify QuAD frameworks when there is a mismatch, due to incompleteness of information, between their intuition and the results of the DF-QuAD algorithm (e.g. the ranking of answers derived from it).

2 Background

A QuAD (*Quantitative Argumentation Debate*) framework (Baroni et al. 2015) is a 5-tuple $\langle \mathcal{A}, \mathcal{C}, \mathcal{P}, \mathcal{R}, \mathcal{BS} \rangle$ such that $\mathcal{A}$ is a finite set of *answer arguments*; $\mathcal{C}$ is a finite set of *con arguments*; $\mathcal{P}$ is a finite set of *pro arguments*; the sets $\mathcal{A}$, $\mathcal{C}$ and $\mathcal{P}$ are pairwise disjoint³; $\mathcal{R} \subseteq (\mathcal{C} \cup \mathcal{P}) \times (\mathcal{A} \cup \mathcal{C} \cup \mathcal{P})$ is an acyclic binary relation; for $\mathbb{I} = [0, 1]$, $\mathcal{BS} : (\mathcal{A} \cup \mathcal{C} \cup \mathcal{P}) \rightarrow \mathbb{I}$ is a total function: $\mathcal{BS}(a)$ is the *base score* of a .

Given any $\mathcal{BS}$, Figure 1 corresponds to a QuAD framework with $\mathcal{A} = \{A1, A2\}$, $\mathcal{C} = \{C1, C2, C3, C4\}$, $\mathcal{P} = \{P1, P2, P3\}$ and $\mathcal{R} = \{(P1, A1), (P2, A1), (C2, P2),$

$(C3, P2), (C4, P2), (P3, A2), (C1, A2)\}$.⁴ Throughout the paper we will refer to this framework, with $\mathcal{BS}(a) = 0.5$ for all $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$, as $\mathcal{Q}_0$. Unless specified otherwise, we will assume as given a generic QuAD framework $\mathcal{Q} = \langle \mathcal{A}, \mathcal{C}, \mathcal{P}, \mathcal{R}, \mathcal{BS} \rangle$.

Pro and con arguments determine attackers and supporters of arguments they are in relation with. Formally, for $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$, the set of *attackers* of a is $\mathcal{R}^-(a) = \{b \in \mathcal{C} \mid (b, a) \in \mathcal{R}\}$ and the set of *supporters* of a is $\mathcal{R}^+(a) = \{b \in \mathcal{P} \mid (b, a) \in \mathcal{R}\}$. In $\mathcal{Q}_0$, $\mathcal{R}^-(P2) = \{C2, C3, C4\}$ and $\mathcal{R}^+(A2) = \{P3\}$.

Due to the acyclicity requirement, QuAD frameworks amount to sets of trees, with answers as roots. Moreover, each argument is the root of a (sub-)tree. For any argument $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$, we will use T_a to denote the tree with root a such that, for any node b in T_a , the children of b are $\mathcal{R}^-(b) \cup \mathcal{R}^+(b)$. For example, for $\mathcal{Q}_0$, T_{A2} has root A2 with children $P3, C1$; these are leaves of T_{A2} .

The QuAD algorithm determines the strength of arguments from their base scores and the aggregated strength of their attackers and supporters. The basic functions $f_{att} : \mathbb{I} \times \mathbb{I} \rightarrow \mathbb{I}$ and $f_{supp} : \mathbb{I} \times \mathbb{I} \rightarrow \mathbb{I}$, defined as follows, for $v_0, v \in \mathbb{I}$:

$$f_{att}(v_0, v) = v_0 - v_0 \cdot v = v_0 \cdot (1 - v) \quad (1)$$

$$f_{supp}(v_0, v) = v_0 + (1 - v_0) \cdot v = v_0 + v - v_0 \cdot v \quad (2)$$

determine how the strength of an argument is obtained by modifying its base score v_0 , in the presence of a single attacker or supporter, respectively, of strength v . Then, functions $\mathcal{F}_x$, for x either *att* or *supp*, handle sequences $S = (s_1, \dots, s_k) \in \mathbb{I}^* \triangleq \bigcup_{i \geq 0} \mathbb{I}^i$ of strengths of attackers or supporters. A sequence is said to be *ineffective* if it is empty or all its elements are 0. Formally, the set of ineffective sequences is $\mathcal{Z} = \bigcup_{i \geq 0} \{0\}^i$ (trivially $\mathcal{Z} \subset \mathbb{I}^*$). We refer to sequences in $\mathbb{I}^* \setminus \mathcal{Z}$ as *non-ineffective*. Also, for any sequence $S = (s_1, \dots, s_k) \in \mathbb{I}^*$ where $v \in \mathbb{I}$, $S \cup (v)$ denotes the sequence $(s_1, \dots, s_k, v) \in \mathbb{I}^{k+1}$.

$\mathcal{F}_x : \mathbb{I} \times \mathbb{I}^* \rightarrow (\mathbb{I} \cup \{\text{nil}\})$ is defined differently for non-ineffective and ineffective sequences, as follows, for $v_0 \in \mathbb{I}$:

- for a non-ineffective sequence $S = (v_1, \dots, v_n) \in \mathbb{I}^* \setminus \mathcal{Z}$:
if $n = 1$, $\mathcal{F}_x(v_0, S) = f_x(v_0, v_1)$ (3)
if $n > 1$, $\mathcal{F}_x(v_0, S) = f_x(\mathcal{F}_x(v_0, (v_1, \dots, v_{n-1})), v_n)$ (4)
- for an ineffective sequence $Z \in \mathcal{Z}$:

$$\mathcal{F}_{att}(v_0, Z) = \mathcal{F}_{supp}(v_0, Z) = \text{nil} \quad (5)$$

We will use the following properties (Baroni et al. 2015):

$$\mathcal{F}_{att}(v_0, S) = v_0 \cdot \prod_{i=1}^n (1 - v_i) \quad (6)$$

$$\mathcal{F}_{supp}(v_0, S) = 1 - (1 - v_0) \cdot \prod_{i=1}^n (1 - v_i) \quad (7)$$

Functions $\mathcal{F}_x$ define how to (recursively) aggregate (the strength of) attackers and supporters separately. Function

³We impose this requirement without loss of generality, as if an argument attacks and supports different arguments, it can be duplicated to the same effect, in such a way that one duplicate argument attacks only and the other supports only.

⁴Issues are present in IBIS graphs but not in QuAD frameworks. Implicitly, each QuAD framework has a single issue, to which all answers are connected.

$g : \mathbb{I} \times (\mathbb{I} \cup \{nil\}) \times (\mathbb{I} \cup \{nil\}) \rightarrow \mathbb{I}$, given below, defines how to aggregate (the strength of) attackers and supporters combined, for $v_0, v_a, v_s \in \mathbb{I}$:

$$g(v_0, v_a, v_s) = v_a \quad \text{if } v_a \neq nil \text{ and } v_s = nil \quad (8)$$

$$g(v_0, v_a, v_s) = v_s \quad \text{if } v_a = nil \text{ and } v_s \neq nil \quad (9)$$

$$g(v_0, v_a, v_s) = v_0 \quad \text{if } v_s = v_a = nil \quad (10)$$

$$g(v_0, v_a, v_s) = \frac{v_a + v_s}{2} \quad \text{otherwise} \quad (11)$$

The strength of an argument is the output of the *score function* $\mathcal{SF}_1 : \mathcal{A} \cup \mathcal{C} \cup \mathcal{P} \rightarrow \mathbb{I}$, which uses the aggregating function g with inputs of the argument's base score and sequences of strengths of its attackers and supporters, as given by the $\mathcal{F}_x$ functions. Formally, for $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$:

$$\mathcal{SF}_1(a) = g(\mathcal{BS}(a), \mathcal{F}_{att}(\mathcal{BS}(a), SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a))), \mathcal{F}_{supp}(\mathcal{BS}(a), SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)))) \quad (12)$$

where if $(a_1, \dots, a_n)$ is an arbitrary permutation of the $(n \geq 0)$ attackers in $\mathcal{R}^-(a)$, $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)) = (\mathcal{SF}_1(a_1), \dots, \mathcal{SF}_1(a_n))$ (similarly for supporters).

For example, for Q_0 , $\mathcal{SF}_1(C1) = \mathcal{BS}(C1) = 0.5$ by Eq. 10, since $C1$ has no attackers/supporters and $\mathcal{F}_{att}(\mathcal{BS}(C1), Z) = \mathcal{F}_{supp}(\mathcal{BS}(C1), Z) = nil$. Similarly $\mathcal{SF}_1(P3) = 0.5$. For $A2$, with both attackers and supporters, Eqs. 12 and 11 give $\mathcal{SF}_1(A2) = \frac{\mathcal{F}_{att}(\mathcal{BS}(A2), 0.5) + \mathcal{F}_{supp}(\mathcal{BS}(A2), 0.5)}{2}$, which, by Eqs. 3, 2 and 1, gives $\mathcal{SF}_1(A2) = \frac{0.25 + 0.75}{2} = 0.5$.

We will show that our notion of strength admits acceptability under grounded extensions in abstract argumentation (AA) (Dung 1995) as an instance. An AA framework is a pair $\langle \text{Args}, \text{Atts} \rangle$ where Args is a set of arguments and Atts is a binary relation on Args , i.e. $\text{Atts} \subseteq \text{Args} \times \text{Args}$. We will denote the *grounded extension* of an AA framework $\mathcal{AF}$ as $GE(\mathcal{AF})$ (see (Dung 1995) for a definition). To relate our approach to AA, we will use Proposition 5.2 in (Baroni et al. 2015). This states that, for $\mathcal{AF} = \langle \text{Args}, \text{Atts} \rangle$ with Atts acyclic, given the QuAD framework $\langle \emptyset, \text{Args}, \emptyset, \text{Atts}, \mathcal{BS}_{\text{Args}}^1 \rangle$ where $\mathcal{BS}_{\text{Args}}^1 \triangleq \text{Args} \times \{1\}$, for every $a \in \text{Args}$: $\mathcal{SF}_1(a) = 1$ if $a \in GE(\mathcal{AF})$, else $\mathcal{SF}_1(a) = 0$.

3 The DF-QuAD Algorithm

We now define the new DF-QuAD algorithm for calculating the strength of arguments in QuAD frameworks. Like the QuAD algorithm, we need to specify how the strength of attackers and supporters is aggregated separately. Differently from the QuAD algorithm, though, we define a *single* such function:

Definition 1. The *strength aggregation function* is defined as $\mathcal{F} : \mathbb{I}^* \rightarrow \mathbb{I}$, where for $S = (v_1, \dots, v_n) \in \mathbb{I}^*$:

$$\text{if } n = 0 : \mathcal{F}(S) = 0 \quad (13)$$

$$\text{if } n = 1 : \mathcal{F}(S) = v_1 \quad (14)$$

$$\text{if } n = 2 : \mathcal{F}(S) = f(v_1, v_2) \quad (15)$$

$$\text{if } n > 2 : \mathcal{F}(S) = f(\mathcal{F}(v_1, \dots, v_{n-1}), v_n) \quad (16)$$

with the *base function* $f : \mathbb{I} \times \mathbb{I} \rightarrow \mathbb{I}$ defined, for $v_1, v_2 \in \mathbb{I}$, as:

$$f(v_1, v_2) = v_1 + (1 - v_1) \cdot v_2 = v_1 + v_2 - v_1 \cdot v_2 \quad (17)$$

The base function is the building block to handle sequences of strengths of attackers or supporters by f . It is defined as f_{supp} in the QuAD algorithm (see Eq. 2).

The strength aggregation function $\mathcal{F}$ can be given an equivalent non-recursive formulation:

Lemma 1. For any $S = (v_1, \dots, v_n) \in \mathbb{I}^*$:

$$\mathcal{F}(S) = 1 - \prod_{i=1}^n (1 - v_i) \quad (18)$$

Proof. By induction on n . For $n = 0$, $\mathcal{F}(S) = 0$ by Eq. 13 and $1 - \prod_{i=1}^0 (1 - v_i) = 1 - 1 = 0$. For $n = 1$, $\mathcal{F}(S) = v_1$ by Eq. 14 and $1 - \prod_{i=1}^1 (1 - v_i) = 1 - (1 - v_1) = v_1$. For $n = 2$, $\mathcal{F}(S) = v_1 + v_2 - v_1 \cdot v_2 = 1 - (1 - v_1) \cdot (1 - v_2)$ by Eqs. 15 and 17 and $1 - \prod_{i=1}^2 (1 - v_i) = 1 - (1 - v_1) \cdot (1 - v_2)$. For the induction step assume that Eq. 18 holds for n , namely $\mathcal{F}(S) = 1 - \prod_{i=1}^n (1 - v_i)$ (induction hypothesis). We need to prove that if $S^+ = S \cup (v_{n+1}) \in \mathbb{I}^*$, then $\mathcal{F}(S^+) = 1 - \prod_{i=1}^{n+1} (1 - v_i)$. By Eq. 16, $\mathcal{F}(S^+) = f(\mathcal{F}(S), v_{n+1})$, which, by the induction hypothesis, gives $\mathcal{F}(S^+) = f([1 - \prod_{i=1}^n (1 - v_i)], v_{n+1})$. Then, by Eq. 17, $\mathcal{F}(S^+) = [1 - \prod_{i=1}^n (1 - v_i)] + v_{n+1} - v_{n+1} \cdot [1 - \prod_{i=1}^n (1 - v_i)] = v_{n+1} + (1 - v_{n+1}) \cdot [1 - \prod_{i=1}^n (1 - v_i)] = v_{n+1} + 1 - v_{n+1} - (1 - v_{n+1}) \cdot \prod_{i=1}^n (1 - v_i) = 1 - (1 - v_{n+1}) \cdot \prod_{i=1}^n (1 - v_i) = 1 - \prod_{i=1}^{n+1} (1 - v_i) \quad \square$

Proposition 1. For any $(v_1, \dots, v_k) \in \mathbb{I}^*$, $\mathcal{F}(v_1, \dots, v_k) \in \mathbb{I}$.

Proof. Directly from Lemma 1. $\square$

When an ineffective sequence is given in input to $\mathcal{F}$, the output is 0, rather than *nil* as in the case of $\mathcal{F}_{att}/\mathcal{F}_{supp}$ in the QuAD algorithm:

Lemma 2. For every $Z \in \mathcal{Z}$, $\mathcal{F}(Z) = 0$.

Proof. Directly from Lemma 1. $\square$

Once the strengths of attacking/supporting arguments against/for (respectively) an argument have been aggregated separately using $\mathcal{F}$, the following function will be used to combine the distance between the two with the base score of the argument, in different ways depending on which, amongst attackers and supporters, is stronger:

Definition 2. The *combination function* is defined as $c : \mathbb{I} \times \mathbb{I} \times \mathbb{I} \rightarrow \mathbb{I}$, where for $v_0, v_a, v_s \in \mathbb{I}$:

$$c(v_0, v_a, v_s) = v_0 - v_0 \cdot |v_s - v_a| \quad \text{if } v_a \geq v_s \quad (19)$$

$$c(v_0, v_a, v_s) = v_0 + (1 - v_0) \cdot |v_s - v_a| \quad \text{if } v_a < v_s \quad (20)$$

The intuition behind this function is that if the attacking strength is larger (respectively, smaller) the base score is decreased (respectively, increased) using the absolute value of the difference between the attacking and supporting strengths to obtain a score that is proportionally closer to 0 (respectively, 1).

The following results provide alternative reformulations of c that will be useful later:

Lemma 3. For any $v_0, v_a, v_s \in \mathbb{I}$:

$$c(v_0, v_a, v_s) = v_0 \quad \text{if } v_s = v_a \quad (21)$$

$$c(v_0, v_a, v_s) = v_0 + (0.5 + \frac{v_s - v_a}{2 \cdot |v_s - v_a|} - v_0) \cdot |v_s - v_a| \quad \text{otherwise} \quad (22)$$

Proof. If $v_a = v_s$, by Eq. 19, $c(v_0, v_a, v_s) = v_0 - v_0 \cdot |0| = v_0$ and Eq. 21 holds. If $v_s > v_a$, Eq. 22 gives $c(v_0, v_a, v_s) = v_0 + (0.5 + \frac{1}{2} - v_0) \cdot |v_s - v_a| = v_0 + (1 - v_0) \cdot |v_s - v_a|$, corresponding to Eq. 20. If $v_a > v_s$, Eq. 22 becomes $c(v_0, v_a, v_s) = v_0 + (0.5 - \frac{1}{2} - v_0) \cdot |v_s - v_a| = v_0 - v_0 \cdot |v_s - v_a|$, corresponding to Eq. 19. $\square$

Lemma 4. For any $v_0, v_a, v_s \in \mathbb{I}$:

$$c(v_0, v_a, v_s) = 1 - \mathcal{F}((1 - v_0), |v_s - v_a|) \quad \text{if } v_a \geq v_s \quad (23)$$

$$c(v_0, v_a, v_s) = \mathcal{F}(v_0, |v_s - v_a|) \quad \text{if } v_a < v_s \quad (24)$$

Proof. Combining Eqs. 23, 15, 17 gives $c(v_0, v_a, v_s) = 1 - [(1 - v_0) + |v_s - v_a| - (1 - v_0) \cdot |v_s - v_a|] = 1 - (1 - v_0 + v_0 \cdot |v_s - v_a|) = v_0 - v_0 \cdot |v_s - v_a|$, which holds by Eq. 19. Combining Eqs. 24, 15, 17 gives $c(v_0, v_a, v_s) = v_0 + |v_s - v_a| - v_0 \cdot |v_s - v_a| = v_0 + (1 - v_0) \cdot |v_s - v_a|$, which holds by Eq. 20. $\square$

Lemma 5. For any $v_0 \in \mathbb{I}$, $Z_1, Z_2 \in \mathcal{Z}$, $c(v_0, \mathcal{F}(Z_1), \mathcal{F}(Z_2)) = v_0$.

Proof. Directly from Lemma 2 and Eq. 21. $\square$

The score function giving the arguments' strength determines the inputs for the combination function as follows:

Definition 3. The *score function* is defined as $\mathcal{SF}_2 : \mathcal{A} \cup \mathcal{C} \cup \mathcal{P} \rightarrow \mathbb{I}$ where, for any $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$:

$$\mathcal{SF}_2(a) = c(\mathcal{BS}(a), \mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a))), \mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^+(a)))) \quad (25)$$

where if $(a_1, \dots, a_n)$ is an arbitrary permutation of the $(n \geq 0)$ attackers in $\mathcal{R}^-(a)$, $\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a)) = (\mathcal{SF}_2(a_1), \dots, \mathcal{SF}_2(a_n))$ (similarly for supporters).

As an illustration, consider the framework $\mathcal{Q}_0$ (see Section 2). For $P1$, with no attackers or supporters, by Eq. 25 and Lemma 5, $\mathcal{SF}_2(P1) = \mathcal{BS}(P1) = 0.5$. Similarly, $\mathcal{SF}_2(P3) = \mathcal{SF}_2(C1) = \mathcal{SF}_2(C2) = \mathcal{SF}_2(C3) = \mathcal{SF}_2(C4) = 0.5$. For $A2$, with a single attacker and a single supporter, Eq. 14 gives both aggregated attacking and supporting strengths as $\mathcal{F}(0.5) = 0.5$. Eqs. 25 and 21 then give $\mathcal{SF}_2(A2) = \mathcal{BS}(A2) = 0.5$. For $P2$, with over two attackers, Eqs. 16 and 17 give $\mathcal{F}(0.5, 0.5, 0.5) = f(f(0.5, 0.5), 0.5) = 0.875$ for the aggregated attacking strength, while Eq. 13 gives $\mathcal{F}(()) = 0$ for the aggregated supporting strength. Given that $v_a > v_s$, Eqs. 25 and 19 give $\mathcal{SF}_2(P2) = 0.5 - 0.5 \cdot |0 - 0.875| = 0.0625$. Finally, for $A1$, with two supporters, Eqs. 15 and 17 give $\mathcal{F}(0.5, 0.0625) = f(0.5, 0.0625) = 0.53125$ for the aggregated supporting strength, while Eq. 13 gives $\mathcal{F}(()) = 0$ for the aggregated attacking strength. Given that $v_s > v_a$, Eqs. 25 and 20 give $\mathcal{SF}_2(A1) = 0.5 + (1 - 0.5) \cdot |0.53125 - 0| = 0.765625$.

4 Properties

We will first prove a number of properties which hold for the DF-QuAD algorithm and whose equivalent holds for the QuAD algorithm. The first property shows the independence of the output of the score function from the order of the input arguments. This is especially important for debates where arguments are added at different times.

Proposition 2. For any sequence $(v_1, \dots, v_k) \in \mathbb{I}^*$, let $(v_{1_i}, \dots, v_{k_i})$ be an arbitrary permutation of $(v_1, \dots, v_k)$. It holds that $\mathcal{F}(v_1, \dots, v_k) = \mathcal{F}(v_{1_i}, \dots, v_{k_i})$.

Proof. Directly from Lemma 1. $\square$

Replacing an attacker (supporter) with one with higher strength does not lower the aggregated attacking (supporting, respectively) strength:

Proposition 3. For any sequence $S = (v_1, \dots, v_l, \dots, v_k) \in \mathbb{I}^*$, $1 \leq l \leq k$, let S^+ be a sequence obtained from S by replacing v_l with some $v_h > v_l$, $v_h \in \mathbb{I}$. Then $\mathcal{F}(S) \leq \mathcal{F}(S^+)$.

Proof. If $k=1$, by Eq. 14, $\mathcal{F}(S)=v_l$ and $\mathcal{F}(S^+)=v_h$. If $k=2$, by Eqs. 15, 17, $\mathcal{F}(S) = v_1 + v_l - v_1 \cdot v_l = v_1 + v_l \cdot (1 - v_1)$ and $\mathcal{F}(S^+) = v_1 + v_h - v_1 \cdot v_h = v_1 + v_h \cdot (1 - v_1)$. If $k>2$, let $S^* = (v_1, \dots, v_{l-1}, v_{l+1}, \dots, v_k)$. By Eqs. 16, 17, $\mathcal{F}(S) = f(\mathcal{F}(S^*), v_l) = \mathcal{F}(S^*) + (1 - \mathcal{F}(S^*)) \cdot v_l$ and $\mathcal{F}(S^+) = f(\mathcal{F}(S^*), v_h) = \mathcal{F}(S^*) + (1 - \mathcal{F}(S^*)) \cdot v_h$. In all cases, since $v_h > v_l$, the lemma holds. $\square$

An analogous result holds when replacing an attacker/supporter with one with a lower strength:

Proposition 4. For any sequence $S = (v_1, \dots, v_h, \dots, v_k) \in \mathbb{I}^*$, $1 \leq h \leq k$, let S^- be a sequence obtained from S by replacing v_h with some $v_l < v_h$, $v_l \in \mathbb{I}$. Then $\mathcal{F}(S) \geq \mathcal{F}(S^-)$.

Proof. Analogous to the proof of Proposition 3. $\square$

Adding an attacker (supporter) will not lower the aggregated attacking (supporting, respectively) strength, and will have no effect if its strength is 0:

Proposition 5. For any $v \in \mathbb{I}$ and $S = (v_1, \dots, v_n) \in \mathbb{I}^*$ for $n \geq 0$, $\mathcal{F}(S \cup (v)) \geq \mathcal{F}(S)$.

Proof. If $n=0$, $\mathcal{F}(S)=0$ by Eq. 13 and $\mathcal{F}(S \cup (v))=v$ by Eq. 14 and $v \geq 0$. If $n=1$, $\mathcal{F}(S)=v_1$ by Eq. 14 and $\mathcal{F}(S \cup (v)) = v_1 + v - (v_1) \cdot v$ by Eqs. 15, 17. If $n \geq 2$, $\mathcal{F}(S \cup (v)) = f(\mathcal{F}(S), v) = \mathcal{F}(S) + v - \mathcal{F}(S) \cdot v$ by Eqs. 16, 17 and $\mathcal{F}(S) + v - \mathcal{F}(S) \cdot v \geq \mathcal{F}(S)$ since, by Proposition 1, $\mathcal{F}(S) \in \mathbb{I}$. $\square$

Proposition 6. For any $S = (v_1, \dots, v_n) \in \mathbb{I}^*$ for $n \geq 0$, $\mathcal{F}(S \cup (0)) = \mathcal{F}(S)$.

Proof. If $n = 0$, $\mathcal{F}(S) = 0$ by Eq. 13 and $\mathcal{F}(S \cup (0)) = 0$ by Eq. 14. If $n = 1$, $\mathcal{F}(S) = v_1$ by Eq. 14 and $\mathcal{F}(S \cup (0)) = v_1 + 0 - v_1 \cdot 0 = v_1$ by Eqs. 15, 17. If $n \geq 2$, $\mathcal{F}(S \cup (0)) = f(\mathcal{F}(S), 0) = \mathcal{F}(S) + 0 - \mathcal{F}(S) \cdot 0 = \mathcal{F}(S)$ by Eqs. 16, 17. $\square$

Adding an attacker/supporter with maximum strength saturates the strength aggregation function:

Proposition 7. For any $S = (v_1, \dots, v_n) \in \mathbb{I}^*$ for $n \geq 0$, $\mathcal{F}(S \cup (1)) = 1$.

Proof. If $n=0$, $\mathcal{F}(S \cup \{1\})=1$ by Eq. 14. If $n=1$, $\mathcal{F}(S \cup \{1\}) = v_1 + 1 - v_1 \cdot 1 = 1$ by Eqs. 15, 17. If $n \geq 2$, $\mathcal{F}(S \cup \{1\}) = f(\mathcal{F}(S), 1) = \mathcal{F}(S) + 1 - \mathcal{F}(S) \cdot 1 = 1$ by Eqs. 16, 17. $\square$

An attacker/supporter with maximum strength is needed for the strength aggregation function to saturate:

Proposition 8. For any $S = (v_1, \dots, v_n) \in \mathbb{I}^*$ for $n \geq 0$, if $\nexists v_i \in S : v_i = 1$ then $\mathcal{F}(S) \neq 1$.

Proof. If $n=0$, by Eq. 13 $\mathcal{F}(S)=0$. If $n \geq 1$, clearly by Eq. 18 for $\mathcal{F}(S)=1$, $\prod_{i=1}^n (1 - v_i) = 0$ and so $\exists v_i \in S : v_i = 1$. $\square$

The extreme values, 0 and 1, can only be achieved as the strength of an argument if they are somehow already present, as follows:

Proposition 9. For any $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$ with $\mathcal{BS}(a) = v_0$, $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a))) = v_a$ and $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^+(a))) = v_s$:

$$\mathcal{SF}_2(a) = 0 \text{ iff } [v_a \geq v_s \wedge v_0 = 0] \vee [v_a = 1 \wedge v_s = 0] \quad (26)$$

$$\mathcal{SF}_2(a) = 1 \text{ iff } [v_a \leq v_s \wedge v_0 = 1] \vee [v_a = 0 \wedge v_s = 1] \quad (27)$$

Proof. The if-half of (26) holds by Eq. 19. To prove the only-if-half of (26) we consider all cases of $c(v_0, v_a, v_s) = 0$ in Eqs. 19, 20. If $v_a < v_s$, by Eq. 20, $v_0 + |v_s - v_a| = v_0 \cdot |v_s - v_a|$. Given that $v_0, |v_s - v_a| \in \mathbb{I}$, this gives $v_0 = 0$ and $|v_s - v_a| = 0$, and thus $v_s = v_a$. But we have assumed that $v_a < v_s$, so Eq. 20 can not apply and necessarily $v_a \geq v_s$. Then, by Eq. 19, $c(v_0, v_a, v_s) = 0$ implies $v_0 = v_0 \cdot |v_s - v_a|$. This equation admits two solutions: $v_0 = 0$, which, along with $v_a \geq v_s$ makes up the first disjunct of the right-hand side of (26), and $|v_s - v_a| = 1$. Since $v_a \geq v_s$, then $v_a = 1$ and $v_s = 0$, which is the second disjunct of the right-hand side of (26).

The proof of (27) is similar (omitted for lack of space). $\square$

The next two results prove an important symmetry between the effect of attackers/supporters and the distance with respect to the two extreme values of 0/1, respectively.

Proposition 10. Let $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$, with $\mathcal{BS}(a) = v_0$, be such that $\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a)) = (v_1, \dots, v_k)$ for some $k \geq 1$ and $\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^+(a)) \in \mathcal{Z}$. Then $\frac{\mathcal{SF}_2(a)-0}{v_0-0} = \prod_{i=1}^k (1 - v_i)$.

Proof. By Eqs. 25, 19, and 18, $\mathcal{SF}_2(a) = v_0 - v_0 \cdot [1 - \prod_{i=1}^k (1 - v_i)]$. This can be rearranged to give the required equation. $\square$

Proposition 11. Let $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$, with $\mathcal{BS}(a) = v_0$, be such that $\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^+(a)) = (v_1, \dots, v_k)$ for some $k \geq 1$ and $\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a)) \in \mathcal{Z}$. Then $\frac{1-\mathcal{SF}_2(a)}{1-v_0} = \prod_{i=1}^k (1 - v_i)$.

Proof. By Eqs. 25, 20, and 18, $\mathcal{SF}_2(a) = v_0 + (1 - v_0) \cdot [1 - \prod_{i=1}^k (1 - v_i)]$. This can be rearranged to give the required equation. $\square$

It follows from Proposition 5 that adding an attacker (supporter) will not increase (decrease) an argument's strength, as implied by the next two results (respectively):

Proposition 12. For any $S_1, S_2 \in \mathbb{I}^*$ and $v, v_0 \in \mathbb{I}$:

$$c(v_0, \mathcal{F}(S_1), \mathcal{F}(S_2)) \geq c(v_0, \mathcal{F}(S_1 \cup \{v\}), \mathcal{F}(S_2))$$

Proposition 13. For any $S_1, S_2 \in \mathbb{I}^*$ and $v, v_0 \in \mathbb{I}$:

$$c(v_0, \mathcal{F}(S_1), \mathcal{F}(S_2)) \leq c(v_0, \mathcal{F}(S_1), \mathcal{F}(S_2 \cup \{v\}))$$

Thus far we have proven properties that are already known to hold also for the original QuAD algorithm. We will now describe some properties of the DF-QuAD algorithm which have no counterparts in the QuAD algorithm. The first such property extends Propositions 10 and 11 to cases when both attackers and supporters are present and are non-ineffective:

Lemma 6. For any $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$, with $\mathcal{BS}(a) = v_0$, $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a))) = v_a$, $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^+(a))) = v_s$:

$$\text{if } v_a \geq v_s, \frac{\mathcal{SF}_2(a)-0}{v_0-0} = 1 - |v_s - v_a|, \text{ and}$$

$$\text{if } v_s > v_a, \frac{1-\mathcal{SF}_2(a)}{1-v_0} = 1 - |v_s - v_a|.$$

Proof. By Eqs. 19 and 25 $\mathcal{SF}_2(a) = v_0 - v_0 \cdot |v_s - v_a| = v_0 \cdot (1 - |v_s - v_a|)$, which can be rearranged to give $\frac{\mathcal{SF}_2(a)-0}{v_0-0} = 1 - |v_s - v_a|$. By Eqs. 20 and 25 $\mathcal{SF}_2(a) = v_0 + (1 - v_0) \cdot |v_s - v_a|$, therefore $1 - \mathcal{SF}_2(a) = 1 - [v_0 + (1 - v_0) \cdot |v_s - v_a|] = 1 - v_0 - |v_s - v_a| + v_0 \cdot |v_s - v_a| = (1 - v_0) \cdot (1 - |v_s - v_a|)$, which can be rearranged to give $\frac{1-\mathcal{SF}_2(a)}{1-v_0} = 1 - |v_s - v_a|$. $\square$

Lemma 6 leads directly to Proposition 14 (15), stating that if the aggregated attacking (supporting, respectively) strength is not lower than the aggregated supporting (attacking, respectively) strength, the argument's strength will be not higher (not lower, respectively) than the base score. This is intuitive in many applications where the base score is a sort of starting point, which is moved in a positive or negative direction depending on whether the aggregated attacking or supporting strength is greater.

Proposition 14. For any $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$ with $\mathcal{BS}(a) = v_0$, $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a))) = v_a$ and $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^+(a))) = v_s$ where $v_a \geq v_s$, $\mathcal{SF}_2(a) \leq v_0$.

Proof. Directly from Lemma 6. $\square$

This result does not hold for $\mathcal{SF}_1$, as shown next:

Example 1. Consider a QuAD framework with an answer argument $A1$ with $\mathcal{BS}(A1) = 0.1$, a con argument $C1$ with $\mathcal{SF}_1(C1) = 0.5$ and a pro argument $P1$ with $\mathcal{SF}_1(P1) = 0.4$, such that $(C1, A1), (P1, A1) \in \mathcal{R}$. Here, the aggregated attacking strength is greater than the aggregated supporting strength, yet $\mathcal{SF}_1(A1) = 0.255$ and $\mathcal{SF}_1(A1) > \mathcal{BS}(A1)$.

Proposition 15. For any $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$ with $\mathcal{BS}(a) = v_0$, $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^-(a))) = v_a$ and $\mathcal{F}(\text{SEQ}_{\mathcal{SF}_2}(\mathcal{R}^+(a))) = v_s$ where $v_s > v_a$, $\mathcal{SF}_2(a) \geq v_0$.

Proof. Directly from Lemma 6. $\square$

This result does not hold for $\mathcal{SF}_1$, as shown next:

Example 2. Consider the QuAD framework in Example 1 but with $\mathcal{BS}(A1) = 0.9$, $\mathcal{SF}_1(C1) = 0.4$ and $\mathcal{SF}_1(P1) = 0.5$. Here, the aggregated attacking strength is lower than the aggregated supporting strength, yet $\mathcal{SF}_1(A1) = 0.745$ and $\mathcal{SF}_1(A1) < \mathcal{BS}(A1)$.

The following main result expresses the discontinuity-freeness of the DF-QuAD algorithm:

Theorem 1. For any $S_1, S_2 \in \mathbb{I}^*$ and $\epsilon, v_0 \in \mathbb{I}$:

$$\lim_{\epsilon \rightarrow 0^+} c(v_0, \mathcal{F}(S_1 \cup (\epsilon)), \mathcal{F}(S_2)) = c(v_0, \mathcal{F}(S_1), \mathcal{F}(S_2)) \quad (28)$$

$$\lim_{\epsilon \rightarrow 0^+} c(v_0, \mathcal{F}(S_1), \mathcal{F}(S_2 \cup (\epsilon))) = c(v_0, \mathcal{F}(S_1), \mathcal{F}(S_2)) \quad (29)$$

Proof. First we prove that for any generic sequence S of n elements, $\lim_{\epsilon \rightarrow 0^+} \mathcal{F}(S \cup (\epsilon)) = \mathcal{F}(S)$. If $n = 0$, $\mathcal{F}(S) = 0$ by Eq. 13 and $\mathcal{F}(S \cup (\epsilon)) = \epsilon$ by Eq. 14, thus the statement holds. If $n = 1$, $\mathcal{F}(S) = v_1$ by Eq. 14 and $\mathcal{F}(S \cup (\epsilon)) = v_1 + \epsilon - v_1 \cdot \epsilon$ by Eqs. 15, 17, thus the statement holds. If $n \geq 2$, $\mathcal{F}(S \cup (\epsilon)) = f(\mathcal{F}(S), \epsilon) = \mathcal{F}(S) + \epsilon - \mathcal{F}(S) \cdot \epsilon$ by Eqs. 16, 17, which approaches $\mathcal{F}(S)$ as $\epsilon \rightarrow 0^+$. So, for all $n \geq 0$, the statement holds.

To prove Eq. 28, let $\mathcal{F}(S_1) = v_a$, $\mathcal{F}(S_2) = v_s$ and $\mathcal{F}(S_1 \cup (\epsilon)) = v'_a$. We have already proved that $\lim_{\epsilon \rightarrow 0^+} v'_a = v_a$. If $v_a \geq v_s$ then as $\epsilon \rightarrow 0^+$, $v'_a \geq v_s$. Therefore, by Eq. 19, $c(v_0, v_a, v_s) = v_0 - v_0 \cdot |v_s - v_a|$ and $c(v_0, v'_a, v_s) = v_0 - v_0 \cdot |v_s - v'_a|$ so $\lim_{\epsilon \rightarrow 0^+} c(v_0, v'_a, v_s) = c(v_0, v_a, v_s)$. If $v_a < v_s$, by Eq. 20, $c(v_0, v_a, v_s) = v_0 + (1 - v_0) \cdot |v_s - v_a|$ and $c(v_0, v'_a, v_s) = v_0 - v_0 \cdot |v_s - v'_a|$ and, once again, $\lim_{\epsilon \rightarrow 0^+} c(v_0, v'_a, v_s) = c(v_0, v_a, v_s)$.

To prove Eq. 29, let $\mathcal{F}(S_2 \cup (\epsilon)) = v'_s$, so $\lim_{\epsilon \rightarrow 0^+} v'_s = v_s$. If $v_s \leq v_a$ then as $\epsilon \rightarrow 0^+$, $v'_s \leq v_a$. Therefore, by Eq. 19, $c(v_0, v_a, v_s) = v_0 - v_0 \cdot |v_s - v_a|$ and $c(v_0, v_a, v'_s) = v_0 - v_0 \cdot |v'_s - v_a|$ so $\lim_{\epsilon \rightarrow 0^+} c(v_0, v_a, v'_s) = c(v_0, v_a, v_s)$. If $v_a < v_s$, by Eq. 20, $c(v_0, v_a, v_s) = v_0 + (1 - v_0) \cdot |v_s - v_a|$ and $c(v_0, v_a, v'_s) = v_0 - v_0 \cdot |v'_s - v_a|$ and, once again, $\lim_{\epsilon \rightarrow 0^+} c(v_0, v_a, v'_s) = c(v_0, v_a, v_s)$. $\square$

This result does not hold for $\mathcal{SF}_1$, as shown next:

Example 3. Consider a QuAD framework with two answer arguments, $A1$ and $A2$, with $\mathcal{BS}(A1) = \mathcal{BS}(A2) = 0.5$, and two pro arguments $P1$ and $P2$, with $\mathcal{BS}(P1) = \mathcal{BS}(P2) = 0.9$, supporting $A1$ and $A2$ respectively. Let $C1$ be a con argument attacking $A1$, with $\mathcal{BS}(C1) = 0.1$. Here, $\mathcal{SF}_2(A2) = \mathcal{SF}_1(A2) = 0.95$ but $\mathcal{SF}_2(A1) = 0.9$ and $\mathcal{SF}_1(A1) = 0.7$. If we reduce $\mathcal{BS}(C1)$ so that it approaches zero, $\mathcal{SF}_2(A1)$ will approach 0.95 (namely the same as $\mathcal{SF}_2(A2) = \mathcal{SF}_1(A2)$) while $\mathcal{SF}_1(A1)$ will approach 0.725.

5 Comparison with Other Frameworks

The DF-QuAD algorithm gives the same result as the QuAD algorithm when $\mathcal{SF}_1$ equals either of the extreme values.

Theorem 2. For any $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$:

$$\mathcal{SF}_2(a) = 0 \quad \text{if } \mathcal{SF}_1(a) = 0 \quad (30)$$

$$\mathcal{SF}_2(a) = 1 \quad \text{if } \mathcal{SF}_1(a) = 1 \quad (31)$$

Proof. The lengthy but straightforward proof is omitted due to space limitations. $\square$

The reverse of this theorem does not hold:

Example 4. Consider a QuAD framework with two answer arguments $A1$ and $A2$, where $\mathcal{BS}(A1) = 1$, $\mathcal{BS}(A2) = 0$. $A1$ is attacked by one con argument $C1$, with $\mathcal{BS}(C1) = x$,

and supported by one pro argument $P1$, with $\mathcal{BS}(P1) = y$, with $0 < x < y < 1$. $A2$ is attacked by one con argument $C2$, with $\mathcal{BS}(C2) = y$, and supported by one pro argument $P2$, with $\mathcal{BS}(P2) = x$. Here, $\mathcal{SF}_2(A1) = 1$ while $\mathcal{SF}_1(A1) = \frac{2-x}{2} < 1$ and $\mathcal{SF}_2(A2) = 0$ while $\mathcal{SF}_1(A2) = \frac{1-x}{2} > 0$.

The example is an instance of the general fact that $\mathcal{SF}_2$ preserves an extreme base score of 1 (respectively, 0) whenever the supporters (attackers) are collectively stronger than attackers (supporters), while this is not the case with $\mathcal{SF}_1$.

We now show that $\mathcal{SF}_2$ and $\mathcal{SF}_1$ give the same strength for arguments where either attackers (or supporters) are ineffective, provided that the scores of the (non-ineffective) supporters (respectively, attackers) are the same.

Theorem 3. For any $a \in \mathcal{A} \cup \mathcal{C} \cup \mathcal{P}$, if (i) $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)) \in \mathcal{Z}$ and $SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)) = SEQ_{\mathcal{SF}_2}(\mathcal{R}^+(a))$ or (ii) $SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)) \in \mathcal{Z}$ and $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)) = SEQ_{\mathcal{SF}_2}(\mathcal{R}^-(a))$ then $\mathcal{SF}_1(a) = \mathcal{SF}_2(a)$.

Proof. A sequence $Z \in \mathcal{Z}$ must be either empty or consist only of zeros. For any set S such that $SEQ_{\mathcal{SF}_1}(S) \in \mathcal{Z}$, if $SEQ_{\mathcal{SF}_1}(S) = ()$, then $S = \{\}$ and $SEQ_{\mathcal{SF}_2}(S) = ()$, while Theorem 2 proves that if S is a non-empty set of arguments then $SEQ_{\mathcal{SF}_1}(S) = SEQ_{\mathcal{SF}_2}(S)$. Therefore if $SEQ_{\mathcal{SF}_1}(S) \in \mathcal{Z}$, $SEQ_{\mathcal{SF}_1}(S) = SEQ_{\mathcal{SF}_2}(S)$.

Now consider the case where both (i) and (ii) hold: $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)), SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)) \in \mathcal{Z}$ (therefore $SEQ_{\mathcal{SF}_2}(\mathcal{R}^-(a)), SEQ_{\mathcal{SF}_2}(\mathcal{R}^+(a)) \in \mathcal{Z}$). By Eqs. 5 and 10 $\mathcal{SF}_1(a) = v_0$ and by Lemma 5 and Eq. 25 $\mathcal{SF}_2(a) = c(v_0, \mathcal{F}(SEQ_{\mathcal{SF}_2}(\mathcal{R}^-(a))), \mathcal{F}(SEQ_{\mathcal{SF}_2}(\mathcal{R}^+(a)))) = v_0$, therefore they are equal.

For the case where (i) only holds, where $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)) \in \mathcal{Z}$ and $SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)) \notin \mathcal{Z}$ (therefore $SEQ_{\mathcal{SF}_2}(\mathcal{R}^-(a)) \in \mathcal{Z}$), Eqs. 9 and 7 show that if $SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)) = (v_1, \dots, v_n)$, $\mathcal{SF}_1(a) = 1 - (1 - v_0) \cdot \prod_{i=1}^n (1 - v_i)$. Combining Eqs. 20 and 18 gives $\mathcal{SF}_2 = v_0 + (1 - v_0) \cdot [1 - \prod_{i=1}^n (1 - v_i)]$ which, taking into account $SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)) = SEQ_{\mathcal{SF}_2}(\mathcal{R}^+(a))$, can be rearranged to give the same result as $\mathcal{SF}_1(a)$.

Finally, for the case where (ii) only holds, when $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)) \notin \mathcal{Z}$ and $SEQ_{\mathcal{SF}_1}(\mathcal{R}^+(a)) \in \mathcal{Z}$ (therefore $SEQ_{\mathcal{SF}_2}(\mathcal{R}^+(a)) \in \mathcal{Z}$), Eqs. 8 and 6 show that if $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)) = (v_1, \dots, v_n)$, $\mathcal{SF}_1(a) = v_0 \cdot \prod_{i=1}^n (1 - v_i)$. Combining Eqs. 19 and 18 for this case gives $\mathcal{SF}_2 = v_0 - v_0 \cdot [1 - \prod_{i=1}^n (1 - v_i)]$ which once again, taking into account $SEQ_{\mathcal{SF}_1}(\mathcal{R}^-(a)) = SEQ_{\mathcal{SF}_2}(\mathcal{R}^-(a))$, is equal to the same result as $\mathcal{SF}_1(a)$. $\square$

Finally, we show the DF-QuAD algorithm's relationship to AA:

Corollary 1. Given an AA framework $\mathcal{Q} = \langle \text{Args}, \text{Atts} \rangle$ such that Atts is acyclic, let $\langle \emptyset, \mathcal{X}, \emptyset, \text{Atts}, \mathcal{BS}_{\mathcal{X}}^1 \rangle$ be a QuAD framework with $\mathcal{BS}_{\mathcal{X}}^1 \triangleq \mathcal{X} \times \{1\}$. Then, for every $a \in \mathcal{X}$: $\mathcal{SF}_2(a) = 1$ if $a \in GE(\mathcal{Q})$, $\mathcal{SF}_2(a) = 0$ otherwise.

Proof. By Theorem 3, when $SEQ_{\mathcal{SF}_2}(\mathcal{R}^+(a)) \in \mathcal{Z}$, $\mathcal{SF}_2(a) = \mathcal{SF}_1(a)$. Therefore, by Proposition 5.2 in (Baroni et al. 2015), the statement holds. $\square$

6 Two Application Scenarios

We have shown that the DF-QuAD algorithm and the QuAD algorithm give the same results in some cases (e.g. for QuAD frameworks encoding AA frameworks) but differ in general. We now consider again the two application scenarios of *engineering design* and *e-democracy*, mentioned in Section 1, to illustrate how the two algorithms may be used to deal with the needs of different applications of IBIS-style debates. In particular, we discuss the behaviour of the two algorithms in successive stages of the debate leading to the IBIS graph in Figure 1. We have given a reading of the nodes of this graph for engineering design already in Section 1. For e-democracy, the reading for the Issue “how to spend a portion of a council’s budget” may be as follows:

- A1=“build a new cycle path”
- A2=“repair current infrastructure”
- P1=“cyclists complain of dangerous roads”
- P2=“a path would enhance the council’s green image”
- P3=“potholes have caused several accidents recently”
- C1=“significant disruptions to traffic would occur”
- C2=“environmentalists are a fraction of the population”
- C3=“recent policies already enhance this green image”
- C4=“donors do not see the environment as a priority”

For both application scenarios, we consider two *stages*: the first generates the IBIS graph of Figure 2, the second gives the IBIS graph of Figure 3. We refer to the QuAD frameworks corresponding to the IBIS graphs⁵ as $\mathcal{Q}_1$, at Stage 1 and $\mathcal{Q}_2$, at Stage 2. In these QuAD frameworks, we assume the following base scores: $\mathcal{BS}(A1) = \mathcal{BS}(A2) = 0.5$, $\mathcal{BS}(P1) = 0.7$, $\mathcal{BS}(P2) = 0.5$ and $\mathcal{BS}(P3) = 0.9$, and $\mathcal{BS}(C1) = 0.2$. In both scenarios, the medium base score for the answer arguments represents a neutral stand towards the two options before the debate. As for the base scores of the other arguments, in the engineering design setting they may reflect the expertise of the user putting those arguments forwards or the importance of the features the arguments refer to (see (Baroni et al. 2015; Aurisicchio et al. 2015) for a discussion), whereas in the e-democracy setting they may reflect (positive and negative) votes by citizens (e.g. as in (Leite and Martins 2011; Evripidou and Toni 2014), see also (Carstens et al. 2015)).

Stage 1. As predicted by Theorem 3 (since for each argument the set of attackers is empty), the two algorithms fully agree at this stage, in that $\mathcal{SF}_1(a) = \mathcal{SF}_2(a)$ for all (answer, pro and con) arguments a in $\mathcal{Q}_1$, the QuAD framework corresponding to the IBIS graph in Figure 2. In particular, all pro arguments have their base score as their strength, no matter which algorithm is used. Moreover,

$$\begin{aligned}\mathcal{SF}_1(A1) &= \mathcal{SF}_2(A1) = 0.925 \\ \mathcal{SF}_1(A2) &= \mathcal{SF}_2(A2) = 0.950.\end{aligned}$$

We believe that for both the engineering design and the e-democracy scenarios, these results are intuitive: both algorithms give a single strong pro-argument more weight than two comparatively weak pro-arguments; as a result, the second answer is (marginally) stronger than the first answer.

⁵These can be obtained straightforwardly, from $\mathcal{Q}_0$ in Figure 1.

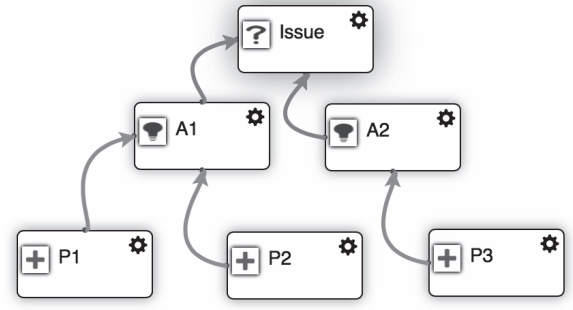


Figure 2: IBIS graph at Stage 1

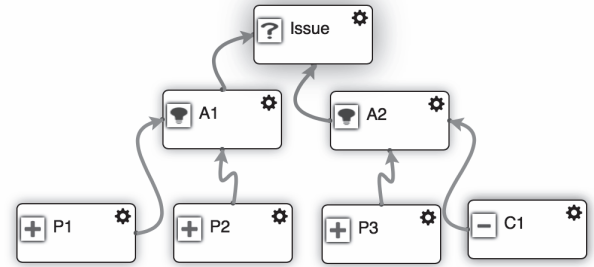


Figure 3: IBIS graph at Stage 2

Stage 2. In Stage 2, a weak con argument, $C1$, is added, attacking $A2$, as shown in Figure 3. In the resulting QuAD framework $\mathcal{Q}_2$, $\mathcal{SF}_1(C1) = \mathcal{SF}_2(C1) = \mathcal{BS}(C1) = 0.2$. As a result of this new attacker against $A2$, its strength decreases, using either algorithm, so that $A2$ becomes weaker than $A1$ and $A1$ becomes the top-ranked answer. However, the original QuAD algorithm reduces this strength to a higher extent than the DF-QuAD algorithm:

$$\begin{aligned}\mathcal{SF}_1(A2) &= 0.675 \\ \mathcal{SF}_2(A2) &= 0.85\end{aligned}$$

The higher drop obtained with $\mathcal{SF}_1$ occurs despite the relatively low strength of $C1$, and is an effect of the discontinuity of $\mathcal{SF}_1$. Instead, the DF-QuAD algorithm affects the strength of $A2$ more proportionally, in line with Theorem 1.

We believe that in the engineering design scenario the DF-QuAD algorithm’s gentler drop in strength is intuitive, and indeed it aligns with conventional matrix-based techniques as for example overviewed in (Aurisicchio et al. 2015).

However, in the e-democracy scenario, getting a disproportionate reduction of strength as a consequence of a first objection (con argument) could be a desirable behaviour. For example, in politics, a solution which is objection free might be preferable to one which raises some (even weak) opposition, even if the latter has stronger positive support.

7 Reverse Engineering

In this section we discuss how, in addition to providing a means for calculating the strength of arguments based on predetermined base scores, the DF-QuAD algorithm could also be used to *engineer* a revised IBIS graph and/or revised

base scores to guarantee a certain ranking between answers.

As an example in the design engineering scenario, a user may “feel”, based on her experience, that the ranking between $A1$ and $A2$ at Stage 2, given by the strength of the answers, is incorrect, and $A2$ really is a superior solution to the issue at hand. Analogously, in the e-democracy example, a politician with her own agenda may want to influence public opinion towards $A2$ rather than $A1$.

If the user were to enquire, at Stage 2, what would be required for $A2$ to once again rank higher than $A1$, the DF-QuAD algorithm could be deployed to provide functionality to determine the exact alterations required in $\mathcal{Q}_2$ for this outcome. Alterations, reversing the ranking of answers, include:

- reducing the strength of $C1$, by either reducing its base score or by adding attackers against it
- reducing the strength of $P1$ and/or $P2$, either by reducing their base score or by adding attackers against them (e.g. Figure 1 represents the addition of attackers to $P2$)
- Increasing the strength of $P3$, either by increasing its base score or by adding supporters for it
- Adding supporters to $A2$
- Adding attackers to $A1$

The above suggestions are concerned with a single level in the graph. If suggestions could include the addition of whole graphs, there would be infinitely many ways to achieve the desired result, even in this relatively simple example.

Here, we focus on providing recommendations amounting to modifying the strength of pro or con arguments to obtain the desired effect – as an intermediate step towards more sophisticated recommendations, that we leave as future work.

As an example, for the strength of $A2$ to be the same as that of $A1$, using the DF-QuAD algorithm as the underlying mechanism for computing strength, the aggregated strength v_a of the attackers against $A2$ (given by $C1$ alone if no modifications to the graph are enforced) would need to be a solution of the following equation: $\mathcal{SF}_2(A2) = 0.5 + (1 - 0.5) \cdot |0.9 - v_a| = 0.925$. Rearranging we obtain:

$$|0.9 - v_a| = \frac{0.925 - 0.5}{1 - 0.5} = 0.85$$

which, since $v_a \in \mathbb{I}$, gives $v_a = 0.05$ as the value that would result in $A2$ having the same strength as $A1$. This, for example, could be achieved by setting $\mathcal{BS}(C1) = 0.05$.

A base score any lower than this would then give $A2$ a higher strength. Other modifications, e.g. addition of supporters for $A2$ or attackers against $C1$, could similarly be obtained by solving equations.

A change to the base score of $C1$, in a design engineering setting, could be meaningful in the following sense. This reverse engineering capability in the DF-QuAD algorithm would be useful in the engineering setting, for example, if the base scores in the debate were related to a parameter. Suppose the base score of “user negligence could lead to losses” is a function of the expected energy or financial cost of the losses. Then, the ability to lower the base score of $C1$ could amount to calculating a limit at which these

losses would cause one answer to be ranked higher than another. This would be extremely useful and potentially provide ground for greater automation in decision support in engineering design.

Note that the QuAD algorithm could not be used to support a similar reverse engineering process, in this case, as no matter how low the strength of $C1$ may be, the strength of $A2$ could not equal that of $A1$, due to the discontinuity. For example, setting $\mathcal{BS}(C1) = 0.05$, would give:

$$\mathcal{SF}_1(A2) = 0.713$$

$$\mathcal{SF}_2(A1) = 0.925$$

Treating this modification of $\mathcal{BS}(C1)$ as a third stage in the process envisaged in Section 6, Figure 4 summarises the strengths of $A1$ and $A2$ at different stages, with the differences between the outputs of the two algorithms highlighted.

Stage	Answer	Strength	
		QuAD	DF-QuAD
1	A1	0.925	0.925
	A2	0.950	0.950
2	A1	0.925	0.925
	A2	0.675	0.850
3	A1	0.925	0.925
	A2	0.713	0.925

Figure 4: Strengths of answers at different stages

8 Related Work

This section details a selection of related work, mainly covering automatic evaluation techniques within argumentation.

The QuAD algorithm (Baroni et al. 2015) introduces QuAD frameworks, used as starting point for the DF-QuAD algorithm. The algorithms differ, however, in that QuAD’s gives a discontinuity which can be counter-intuitive in some applications, as discussed in Sections 6 and 7.

We have used a mapping from AA to QuAD to compare the DF-QuAD algorithm and acceptability in AA. *Bipolar Argumentation Frameworks* (BAFs) (Amgoud et al. 2008) can also be mapped onto QuAD (see (Baroni et al. 2015)). A notable difference between our approach and *gradual valuation* functions for BAFs (Amgoud et al. 2008) is the base score, absent in BAFs. Whether special forms of gradual valuation functions correspond to the output of $\mathcal{SF}_2$, and in particular are discontinuity-free is left for future work.

Social Abstract Argumentation Frameworks (SAFs) (Leite and Martins 2011) use an automatic evaluation algorithm for determining the strength of arguments in AA frameworks extended with a social support for arguments, derived from (positive and negative) votes. The DF-QuAD algorithm is equivalent to the method of (Leite and Martins 2011) for aggregating the strength of attacks, when applied to QuAD frameworks obtained from acyclic AA frameworks and where the base scores result from aggregating votes as in (Leite and Martins 2011). In this sense, the DF-QuAD algorithm can be seen as a generalisation of (Leite and Martins 2011) to the case where support and arbitrary base scores are considered. The method of (Leite and Martins 2011) is

not restricted to acyclic AA frameworks, and could serve as a starting point for generalising the DF-QuAD algorithm to QuAD frameworks where $\mathcal{R}$ is not required to be acyclic.

The social, web-based Q&A debating system www.quaestio-it.com (Evripidou and Toni 2014) is a generalisation of SAFs, using (acyclic) BAFs rather than AA frameworks as their starting point. The algorithm underlying this system aggregates the strength of con arguments as in (Leite and Martins 2011), and in addition aggregates the strength of pro arguments in such a way that the effect of a pro argument does not reduce as the strength of the argument it is supporting approaches 1, as it does using $\mathcal{SF}_1$ and $\mathcal{SF}_2$. Instead, a pro argument increases the strength in the amount that the argument *would* move it towards zero if it were a con argument. Any increment past 1 is reduced down to 1 providing a saturating effect (see also (Carstens et al. 2015)), implying that the equivalent of Proposition 1 holds in a forced manner. This system uses the same aggregating function g as in $\mathcal{SF}_1$, giving rise to the same discontinuity exhibited by the QuAD algorithm.

(Dubois, Fargier, and Bonnefon 2008) propose a qualitative approach to aggregating pro and con arguments for decisions/answers, in which arguments are not assigned a strength value on a continuous scale but only negativity, neutrality or positivity and an importance ranking on an ordered scale. The motivation behind this is the observation that qualitative evaluations are often closer to human capabilities than numerical ones. This work defines a number of *decision rules* which could be used to assess the effectiveness of any strength evaluation algorithms, including ours. For example, the paper dictates that for a set of attacking or supporting arguments to be better than the empty set, the supporting component must be greater than the attacking component. For the DF-QuAD algorithm this equates to $\mathcal{SF}_2(a) > BS(a)$ for an argument a if the supporting strength is larger than the attacking strength, which holds by Proposition 3, Proposition 15 and Lemma 2. However, this does not hold for the QuAD algorithm: if we have an argument a with $BS(a) = 0.1$, an attacker b with $\mathcal{SF}_1(b) = 0.4$ and a supporter c with $\mathcal{SF}_1(c) = 0.2$, then $\mathcal{SF}_1(a) = 0.17$. This means that the score function result is greater than the base score, i.e. the score function result if there are no attacking and supporting arguments, despite the attacking argument being stronger than the supporting one.

Within engineering design there are a range of IBIS-based tools which provide aids for decision-making, all of which differ from our method. The *HERMES* system (Karacapilidis and Papadias 2001) adds numerical weights and constraints representing preferences to the basic IBIS model but this is solely to determine the *activation* of arguments, which comes about when an argument’s supporters are stronger than its attackers. The *EDEN* (Marashi and Davis 2006) system also adds quantitative measures to the IBIS model but these are based on uncertainty, rather than the gradual acceptability in the DF-QuAD algorithm. There are many others within this field, such as the already mentioned design-VUE, which provide some level of IBIS representation or quantitative assessment but none with all of the functionality of the DF-QuAD algorithm. Notable systems for articulated

dialogical argumentation include the Carneades (Gordon, Prakken, and Walton 2007) and Parmenides (Cartwright and Atkinson 2009) platforms, they do not include however any form of numerical evaluation. Both Carneades and QuAD frameworks highlight the importance of dialogical context and use forms of argumentation which depart from the standard Dung-style semantics. Moreover, the notion of proof standard in dialogical argumentation presented in (Gordon and Walton 2009) refers to quantitative weights assigned to arguments by an audience, which are similar to our base scores. In (Gordon and Walton 2009) these weights are not processed to produce final scores but rather used to assess, in a binary way, whether a conclusion meets a proof standard. Considering this alternative perspective within QuAD frameworks is an interesting direction of investigation.

There is much related research on revising argumentation frameworks which would provide other techniques for exploiting the full potential of the reverse engineering direction described in Section 7. For example, (Coste-Marquis et al. 2014) focus on the revision of argumentation systems by modifying the attack relations only, rather than allowing new arguments to be added. This may be useful in a design engineering setting where troublesome constants could be identified. Another recent work is (Baumann and Brewka 2015), which combines abstract argumentation with belief revision to show how AGM-style expansion and revision operators can be defined for the former. The allowance of uncertainty would be very useful in e-democracy since people’s opinions often vary with time. Methods for enforcement within argumentation, as expanded on in (Bisquert et al. 2013), could also be used here, for example, in the form of forcing an answer (not) to be the highest ranked answer in a framework. In an e-democracy setting this could be used to determine the areas in which campaigning needs to be focused for a desired result to be achieved. The investigation of potential relationships between reverse engineering in the DF-QuAD algorithm and these methods is left for future work.

9 Conclusion

The study of algorithms for quantitative decision support in IBIS graphs, a significant instance of argument-based formalism especially oriented to the representation of debates concerning wicked problems, has received up to now limited attention in the literature, though many tools to support the creation and visualization of IBIS graphs are available.

In this paper we have defined a novel algorithm DF-QuAD, using the QuAD algorithm presented in (Baroni et al. 2015) as a starting point. These algorithms share some basic ideas and satisfy some common general requirements. However, the DF-QuAD algorithm features some different properties, and, in particular, is free from the discontinuities shown by the original algorithm in some cases.

Discussing two examples from different domains giving rise to structurally isomorphic IBIS graphs, we have argued that different properties may be intuitively desirable in different contexts. In this spirit, the definition of the DF-QuAD algorithm, rather than being a step in the quest for the *best* decision support algorithm, may be regarded as a contribution towards the construction of a portfolio of algorithms, so

that the most suitable can be selected for each context. The ability of the DF-QuAD algorithm to support reverse engineering of decisions is a notable feature in this respect.

Future work directions include the study of properties of algorithms for argument-based decision support at a more abstract level so as to identify a set of basic domain-independent requirements to be satisfied by any approach and then a set of domain-sensitive features, that may affect algorithm choice in a given context. This kind of investigation is, to our knowledge, a largely open issue and may contribute to fill the gap between the variety of decision making methods and attitudes found in practice and the rigidity of single-algorithm decision support approaches (see e.g. (Aurisicchio et al. 2015) for a comparison and preliminary proposal of integration between the QuAD framework and decision-matrix formalisms). The practical examples in this paper are shown to illustrate the motivations, properties and functionalities of the DF-QuAD algorithm and so a more comprehensive practical evaluation is also left to future work. The actual meanings of the strengths and base scores are open questions, as discussed in (Baroni et al. 2015). In particular, how the base scores of arguments can be obtained in applications will be targeted in the future.

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